

Blind Faith in Text: Evaluating Multi-modal Embedding Robustness in Remote Sensing Land Cover Analysis

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Abstract—Vision-Language Models (VLMs) have been shown to exhibit a “blind faith in text” phenomenon, trusting textual input over visual evidence under modality conflict. This work investigates whether fusion strategy moderates this effect in remote sensing (RS) land cover analysis. We compare a late-fusion dual encoder against a cross-attention fusion encoder, both built on frozen RemoteCLIP encoders with an identical classification head, and evaluate them under three text conditions: matching, corrupted (wrong-class), and irrelevant captions. The dual encoder, which never integrates text into the image representation, is invariant across all conditions (macro F1 = 0.60). The fusion encoder reaches 0.93 macro F1 with matching captions but collapses to 0.05 under corrupted captions, while partially recovering to 0.38 under irrelevant captions. This ordering, which causes catastrophic failure under plausible misinformation but partial recovery under noise, demonstrates that the fusion architecture does not ignore the image, but actively prefers confident yet incorrect text over visual modality. We conclude that fusion strategy strongly moderates the blind-faith effect: cross-attention fusion creates a text-specific vulnerability that late fusion (dual encoder) is immune to.

Index Terms—Vision-Language Models, Remote Sensing, Multi-modal Embeddings, Text Bias, Cross-Attention Fusion, Land Cover Classification

I. INTRODUCTION

VLMs are being adopted to major computer vision domains such as remote sensing for their ability to caption, VQA etc, yet recent work shows that VLMs can exhibit a “blind faith in text” [1]: when textual input conflicts with visual input, the model may trust the text. In RS, where image captions or metadata can be noisy, inaccurate, or automatically generated, this vulnerability is especially consequential. This project asks whether the architectural strategy used to fuse the two modalities, late fusion versus cross-attention fusion, moderates the blind-faith effect. We build both architectures on frozen RemoteCLIP encoders with an identical classification head, isolating fusion strategy as the single variable, and investigate them under matching, corrupted, and irrelevant captions.

II. LITERATURE SURVEY

The transformer architecture [2] and its vision adaptation ViT [3] established the foundation for modern Vision-Language Models (VLMs). After the adoption of transformer

architecture for both models, Li et al. [4] investigates whether we can adopt the multi-modal embeddings on a down-stream task like remote sensing and shows that the multi-modal models can achieve good results, even better than the CNNs, which is highly adopted on the remote sensing domain. There are several attempts to adopt the technology for remote sensing, one example is GeoRSCLIP [5], demonstrating strong zero-shot performance on RS tasks. At a larger scale, EarthGPT [6] and GeoChat [7] extend VLMs to instruction-following for RS applications. There are lots of research for adapting the VLMs on remote-sensing domain, but before that, The gap VLMs possess is not studied enough. Deng et al. [1] identify a “blind faith in text” phenomenon in general VLMs, where models trust textual input over visual evidence under modality conflict, leading to significant performance drops. And VLM models use different fusion techniques to fuse both modalities on a common space, and it leads to 2 different architectures: dual-encoder VLMs and fusion-encoder VLMs [4]. Dual-encoder uses 2 different encoders to generate embeddings for both modalities, and measure similarity with a dot product, while fusion-encoders uses cross-attention to allow deeper relationships. These different architectural choices and their effects on the “blind faith in text” phenomenon is an unexplored study.

III. RESEARCH QUESTION

This project investigates whether the blind-faith-in-text phenomenon occurs in RS land cover analysis and, critically, whether the fusion strategy moderates it. The research question is: *Does caption accuracy affect image-text alignment in RS multi-modal embeddings, and does fusion strategy moderate this effect?*

Phase 1 established feasibility using CLIP on a single grassland image, where a correct caption produced higher cosine similarity (0.2672) than a misleading one (0.2571), confirming that caption quality measurably affects alignment. Phase 2 extended this to the full dataset with RemoteCLIP, measuring cosine similarity and zero-shot classification. Phase 3, presented here, moves from a zero-shot similarity analysis to

a trained architectural comparison, contrasting late fusion and cross-attention fusion under three controlled text conditions.

IV. METHODS

A. Dataset

We use a 10,000-sample subset of ARAS400k [8], an RS segmentation and captioning dataset accepted at a CVPR 2026 workshop. Each sample provides a land-cover composition (percentage per class over seven classes: tree, shrub, grass, crop, built-up, barren, water) and VLM-generated captions. The dominant class of each image is taken as the $\arg\max$ over its composition columns. The class distribution is highly imbalanced (grass 47.0%, tree 30.4%, crop 18.0%; shrub, built-up, barren, and water together under 5%). Data is split 70/30 into train and test sets with stratification on the dominant class.

B. Architecture

Both arms use frozen RemoteCLIP [9] (ViT-B/32) encoders and differ only in how the image representation is formed before an identical classification head (Linear(512, 7)). This single-variable design isolates the effect of fusion strategy.

Dual encoder (late fusion). The pooled image embedding ($\mathbb{R}^{512}$) is passed directly to the head. Text never enters the image representation; the caption cannot influence the prediction.

Fusion encoder (cross-attention). The pre-pool patch sequence ($\mathbb{R}^{50 \times 768}$) is projected to $\mathbb{R}^{50 \times 512}$ and used as queries in a multi-head cross-attention layer, with the text token sequence ($\mathbb{R}^{77 \times 512}$) as keys and values. A residual connection and layer normalization follow; the fused sequence is mean-pooled to $\mathbb{R}^{512}$ and passed to the head. Cross-attention gives text a direct path to modulate the image representation.

C. Training and Evaluation

To avoid recomputing frozen encoders, all image and text embeddings are extracted once and cached. Cache integrity was confirmed before training: zero-shot classification on the cached embeddings reproduced the Phase 2 RemoteCLIP accuracy of 15.95%. Both arms are trained *only on matching captions* using class-weighted cross-entropy (weights inversely proportional to class frequency) to counter imbalance, with the AdamW optimizer. Following Deng et al. [1], each trained model is evaluated under three text conditions: *matching* (the image’s own caption), *corrupted* (a caption sampled from an image of a different dominant class, seed = 42), and *irrelevant* (a domain-unrelated sentence). The matching condition measures the ceiling, corruption measures robustness to confident misinformation, and irrelevance distinguishes genuine text-preference from image-blindness. Because of severe class imbalance, macro F1 is the primary metric, with accuracy reported as a secondary measure.

V. RESULTS

Table I reports macro F1 for both arms across the three text conditions. The dual encoder is invariant: because text never enters its representation, matching, corrupted, and irrelevant captions yield identical results (macro F1 = 0.60, accuracy 77.4%). This invariance serves as a control confirming the experimental setup.

The fusion encoder behaves differently. With matching captions it reaches 0.93 macro F1 (96.9% accuracy), far exceeding the dual baseline—confirming that text, when accurate, adds substantial information. Under corrupted captions it collapses to 0.05 macro F1 (5.1% accuracy), below random chance. Under irrelevant captions it partially recovers to 0.38 macro F1 (49.7% accuracy). Figure 1 visualizes this contrast.

TABLE I
MACRO F1 BY ARCHITECTURE AND TEXT CONDITION (TEST SET, 3,000 IMAGES). THE DUAL ENCODER IS INVARIANT; THE FUSION ENCODER COLLAPSES UNDER CORRUPTION BUT PARTIALLY RECOVERS UNDER IRRELEVANCE.

Architecture	Matching	Corrupted	Irrelevant
Dual (late fusion)	0.60	0.60	0.60
Fusion (cross-attention)	0.93	0.05	0.38

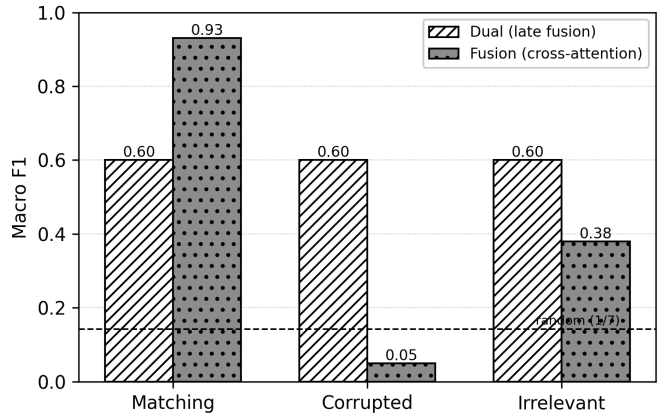


Fig. 1. Macro F1 of the dual (late-fusion) and fusion (cross-attention) encoders under matching, corrupted, and irrelevant captions. The dual encoder is invariant across conditions. The fusion encoder peaks with matching captions, falls below random chance under corruption, and partially recovers under irrelevant captions, indicating an active preference for plausible-but-wrong text over visual evidence. Bars are distinguished by hatch pattern for black-and-white legibility.

The per-class breakdown under corruption (Table II) shows that the fusion model’s predictions follow the misleading captions: the majority classes the captions point away from (tree, grass, crop) drop to near-zero F1, while rare classes that the misleading captions happen to point toward retain marginal recall. The collapse is therefore not random degradation but systematic misdirection by the text.

VI. DISCUSSION

Fusion strategy strongly moderates the blind-faith effect. The dual encoder is structurally not affected by text

TABLE II
FUSION ENCODER PER-CLASS F1 UNDER CORRUPTED CAPTIONS.
PREDICTIONS TRACK THE MISLEADING TEXT RATHER THAN THE IMAGE.

Class	Precision	Recall	F1
Tree	0.01	0.00	0.01
Grass	0.22	0.07	0.10
Crop	0.01	0.01	0.01
Barren	0.05	0.43	0.09
Water	0.06	0.47	0.11

corruption: with no interaction between the text and the image representation, misleading captions cannot affect its predictions. The fusion encoder, by contrast, exhibits a text-specific vulnerability. This directly answers the research question: the architectural choice is not neutral, cross-attention fusion *creates* the conditions for blind faith.

The model prefers text, but does not ignore the image.

A natural concern is that the fusion model simply learned a shortcut, reading the class from the caption and ignoring the image entirely. The irrelevant condition rules this out. If the model ignored vision, irrelevant text would be as damaging as corrupted text, driving performance to the same floor. Instead, performance recovers from 0.05 (corrupted) to 0.38 (irrelevant): when text carries no class signal, the model falls back on the image. The collapse is specific to *confident, plausible but wrong* text. This is precisely the signature of blind faith in text, and it mirrors the finding of Deng et al. [1] that VLMs are less distracted by clearly irrelevant text than by almost relevant text.

Origin of the vulnerability. The fusion model was trained only on aligned (image, correct-caption) pairs and therefore never learned that text can be wrong. Deng et al. attribute blind faith to a training imbalance favouring text; our result is a domain-specific instance of the same mechanism, induced by an all-aligned training regime. This also suggests a cure consistent with their work: augmenting training with corrupted and irrelevant captions should reduce the vulnerability.

Limitations. The rare classes (shrub, built-up, barren, water) have very few test samples (8–58), making their per-class F1 unstable; the macro average is correspondingly sensitive to them. The irrelevant condition draws from a small fixed sentence pool, so its captions repeat; while content is uniformly off-domain, a larger pool would strengthen the claim. Results are from a single run; reporting mean and variance over seeds would explain the magnitude of the effect better.

VII. CONCLUSION AND FUTURE WORK

We compared late-fusion and cross-attention-fusion architectures on RS land cover classification under matching, corrupted, and irrelevant text. Late fusion is invariant to text corruption by construction, while cross-attention fusion, despite achieving the highest accuracy with correct captions, collapses below random under confident misinformation and only partially recovers under noise, demonstrating an active preference for plausible-but-wrong text over visual evidence.

So as a result, fusion strategy strongly moderates the blind-faith-in-text effect.

This study examined a single cross-attention fusion design. Future work could use how *fusion depth* moderates text-trust by comparing early fusion, stacked intermediate cross-attention, and deeper token-level interaction, establishing whether the blind-faith effect scales with the depth of multi-modal integration. Adversarial training with corrupted and irrelevant captions is a further direction for building RS VLMs robust to the phenomenon described in this study, blind-faith in text.

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